

Worksheet 1: Visual and Geometric Patterns

5 Additional problem:

11. How many dots are in the 5th hexagonal number if the sequence starts 1, 6, 15, 28, ...?

- A) 36 B) 45 C) 48 D) 55

Hint: The differences are 5, 9, 13 (increasing by 4 each time). The next difference is 17. $28 + 17 = 45$.

12. If a square number ends in 6, its square root must end in which digit?

- A) 2 or 8 B) 4 or 6 C) 3 or 7 D) 1 or 9

Hint: Look at basic squares: $4 \times 4 = 16$. $6 \times 6 = 36$. Both end in 6.

13. A rectangular dot pattern has 3 rows and 5 columns. If you add one row and one column, how many dots are there now?

- A) 15 B) 20 C) 24 D) 30

Hint: Original is 3×5 . New rectangle is $(3+1) \times (5+1) = 4 \times 6$.

14. What is the next number in the sequence representing the perimeter of squares with side lengths 1, 2, 3, 4...?

- A) 8 B) 16 C) 20 D) 24

Hint: Perimeter = $4 \times \text{side}$. The sequence is 4, 8, 12, 16. The 5th square has perimeter 4×5 .

15. If a cube is made of 27 smaller blocks ($3 \times 3 \times 3$), how many blocks are totally hidden in the centre?

- A) 1 B) 3 C) 8 D) 9

Hint: To find the hidden centre of a $3 \times 3 \times 3$ cube, subtract the outer layer: $(3-2) \times (3-2) \times (3-2) = 1 \times 1 \times 1$.